



Password Strength Checker

A Python Desktop Security Tool

Version: 1.0.0

Author: Abdelrahman Ashraf

GitHub: github.com/G1H6-bit/password-checker

LinkedIn: [linkedin.com/in/abdelrahman-ashraf-39169035a](https://www.linkedin.com/in/abdelrahman-ashraf-39169035a)

Date: June 2026

1. Project Overview

The Password Strength Checker is a Python desktop application that evaluates the security level of any password using real-world cybersecurity rules. It provides instant visual feedback, a strength score, and actionable improvement suggestions — all through a clean graphical interface.

This tool was built to bridge the gap between complex security guidelines and everyday users. Instead of relying on vague descriptions, it breaks down exactly why a password is weak and what steps will make it stronger.

2. Project Information

Language	Python 3.x
GUI Library	Tkinter (built-in — no installation required)
Category	Cybersecurity / Desktop Application
Difficulty	Beginner–Intermediate
Dependencies	None — runs with standard Python 3 only
Platform	Windows / macOS / Linux

3. Key Features

The application delivers a complete password evaluation experience:

- **Real-time strength scoring on a scale of 0 to 5**
- Color-coded feedback — Red (Weak), Orange (Medium), Green (Strong)
- Detection of commonly leaked passwords such as '123456' and 'password'
- Personalized improvement suggestions tailored to each password
- Show / Hide password toggle for user comfort
- Press Enter to check — no mouse click required
- Lightweight — no external libraries or internet connection needed

4. Security Rules Engine

The tool evaluates passwords against five core security rules used in real production systems:

Check	Rule	Requirement
✓ Length	Minimum character count	At least 6 characters (10+ recommended)
✓ Uppercase	Capital letters present	At least one character from A–Z
✓ Lowercase	Small letters present	At least one character from a–z
✓ Numbers	Digits included	At least one digit from 0–9
✓ Symbols	Special characters present	At least one from !@#\$%^&*...

5. How to Run

Follow these steps to set up and run the application:

Step 1 — Prerequisites

- Python 3.x must be installed on your machine
- No additional packages required

Step 2 — Clone the Repository

```
git clone https://github.com/G1H6-bit/password-checker.git
cd password-checker
```

Step 3 — Run the Application

```
python password_checker.py
```

6. How It Works — Logic Flow

When a user types a password and presses Enter or the Check button, the application:

- Step 1 — Checks the password against the list of commonly leaked passwords
- Step 2 — Evaluates each of the 5 security rules independently
- Step 3 — Calculates a total score from 0 (all rules failed) to 5 (all rules passed)

- Step 4 — Maps the score to a label: Weak (0–2), Medium (3–4), Strong (5)
- Step 5 — Updates the UI with the correct color, label, and specific tips

7. What I Learned

Building this project developed key skills in cybersecurity and software development:

- Understanding real password security standards used in industry
- Building desktop GUIs with Python's Tkinter library
- Writing clean, rule-based validation logic
- Applying UX thinking to security tools — making complex rules easy to understand
- Working with common password breach datasets

8. Future Improvements

- Integration with the HaveIBeenPwned API for live breach checking
- Password strength history chart to track improvement
- Password generator feature based on selected strength level
- Dark mode UI theme

9. License & Author

This project is released under the MIT License — free to use, modify, and distribute.

Author	Abdelrahman Ashraf
---------------	--------------------

GitHub	github.com/G1H6-bit
---------------	---

LinkedIn	linkedin.com/in/abdelrahman-ashraf-39169035a
-----------------	---